The pathway heatmap

The measured values behind one Gene Ontology term, gene by gene and sample by sample, in one square.

AT A GLANCE

Where	Statistics → Pathway heatmap
Data levels	Any uploaded level that carries gene names
Needs	Gene names in the dataset. The term list comes from the bundled human Gene Ontology database and is available without running anything first.
One row	One gene of the term, as the median of every dataset row whose annotation names it.
Two value types	Z-score across the displayed samples, or Intensity as the matrix holds it.
Table	None. The page produces the figure only.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

An enrichment result says that a term is over-represented in a list. It says nothing about whether the genes behind it actually move together, or in which samples — a term can be called enriched on a set of genes that go up, down and nowhere. This page puts those genes back on the data: one row per gene of the term, one column per sample, so that a coherent block and a scattering of unrelated rows look different.

Because the term list is the whole bundled database rather than the output of a run, the page also works the other way round: pick a pathway you already care about and look at it, without an enrichment result at all.

Reading it

- One cell: one gene of the term in one sample. Missing values are left uncoloured rather than filled.
- One row: one gene named by the term. Its value is the *median* across every row of the dataset whose gene annotation contains that gene — one row for most genes, several where a protein group or a phosphoprotein names it.
- One column: one sample of a selected condition, labelled condition_n, with the sample identifier on a second line when **Toggle ID** is on.
- The colourbar. Under Z-score it is centred on each row's own mean and measured in that row's own standard deviations; under Intensity it is the matrix's own unit, named in the label.

A GO term can contain genes with different responses; coherent colour bands are not a requirement for a valid enrichment result. The term may describe a cellular component or molecular function rather than a signalling pathway. Check how many annotated genes were measured, which contributed to the query, and which move in different directions. Turn **Remove empty genes** off to inspect coverage. The heatmap itself is not an enrichment test and cannot establish pathway activation or inhibition.

> **EASY TO GET WRONG**
>
> Z-score is computed per row, so every row is stretched to fill the same colour range whatever its real spread. A gene that varies by a factor of ten and one that varies in the third decimal come out looking equally variable, and a row with no spread at all is blanked entirely rather than drawn flat. Use Z-score to compare the *shape* of the rows and Intensity to compare their level; the title says which is on screen.

2 Controls

Controls on this page

Control	What it does	Default	Where
Dataset level	Which dataset to draw. Only levels carrying a gene-name column can be used.	Protein	page

Controls on this page (continued)

Control	What it does	Default	Where
Pathway	The term to draw, chosen from every term description in the bundled Gene Ontology database — 22,074 of them, in alphabetical order.	the first term in the list	page
Value type	Z-score standardises each row across the displayed samples; Intensity draws the values as the matrix holds them.	Z-score	page
Data source	Which matrix to read: Data, Filtered Data or Imputed Data, whichever exist.	the most processed matrix available	page
Conditions	Which conditions supply the columns. Samples are drawn in annotation order within the selection.	all conditions	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure.	20, 12, 300	page
Show gene labels	Prints the gene name beside each row. Turn it off for a term of several hundred genes.	on	page
Toggle ID	Adds the sample identifier under each column label.	off	page
Remove empty genes	Drops rows for genes of the term that no dataset row names — the usual case for most of a large term. Off, they are drawn as empty rows, which is the honest picture of how much of the term was measured.	on	page
Show collapsed row count (n)	Appends (n=k) to each gene label, where *k* is how many dataset rows were collapsed onto it. Off by default: for most genes the answer is 1.	off	page
Cluster rows, Cluster columns	Reorders rows or columns by average linkage on euclidean distances. Both require complete values.	off	page

In the plot appearance panel

Control	What it does	Default	Where
Colour scale	The gradient the cells are drawn in.	plasma	appearance
Title, axis labels, ticks, font, frame, grid, background	As for every other plot; described in Part XI.	the plot's own	appearance

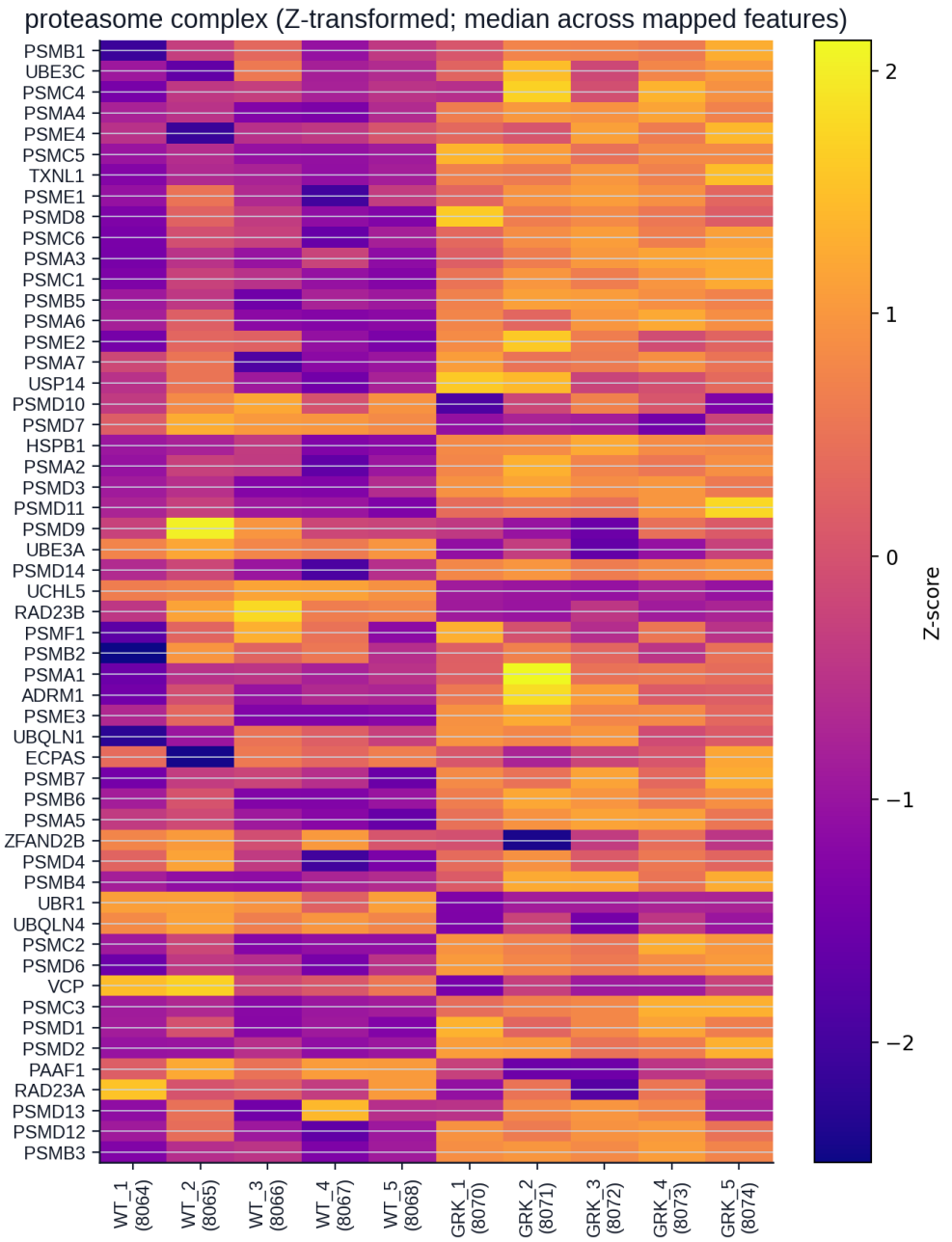


Figure 1.1. proteasome complex at protein level on the example project, Z-scored: one row per gene of the term that the dataset quantified, one column per sample, the five WT samples then the five GRK ones. The block of colder colours on the left against warmer ones on the right is the whole term moving together between the two conditions — which is what this plot exists to show, and what no single volcano point says.

WORTH KNOWING

The appearance panel opens with **Ctrl** + **Shift** + **E**.

3 How it is computed

FROM TERM TO ROWS

The chosen term is matched in the database by its description, ignoring case, and its gene list is read. Those genes, in the term's own order and with repeats removed case-insensitively, are the rows of the figure.

Each dataset row contributes to a gene when its gene annotation contains that gene as one of its semicolon-separated tokens, matched without regard to case. A token repeated inside one annotation still counts as one contribution. Where several dataset rows map to the same gene — a protein group, or several phosphoproteins of one gene — the row's value in each sample is the **median** of theirs, over the values that are present. A gene of the term that no dataset row names becomes a row of NA, and is removed when **Remove empty genes** is on.

THE COLOUR

Under Intensity the numbers are drawn as they stand, and the colourbar takes the matrix's unit — log2 Intensity, or log2 Ratio to Site Median for phosphoprotein data collapsed by the robust site median.

Under Z-score each row i is standardised across the displayed samples only:

$$z_{ij} = \frac{x_{ij} - \bar{x}_i}{s_i},$$

with $\bar{x}_i$ and s_i the mean and standard deviation of that row over the selected samples, both taken over the values that are present. A row whose standard deviation is zero or undefined has no meaningful z-score and is blanked and then dropped rather than divided by zero. The scores therefore change when the condition selection changes: two heatmaps of the same term over different conditions are not comparable cell for cell.

ORDERING

Without clustering the rows are in the term's own gene order and the columns in

annotation order. **Cluster rows** and **Cluster columns** each replace that with the leaf order of an average-linkage tree on euclidean distances, computed on the matrix as drawn — so on z-scores under Z-score and on the values themselves under Intensity. Clustering cannot be computed through a missing value, so when either switch is on the rows that are not observed in every displayed sample are dropped first. On a matrix with scattered missing values that can remove most of the term, which is why a notice appears beside the two switches whenever they are used on anything other than an imputed matrix. Nothing is imputed for you.

Edge cases

- A term none of whose genes appear in the dataset produces no figure and an error saying so.
- With **Remove empty genes** off and few genes matched, the figure is mostly blank: that is the measured coverage of the term, not a fault.
- Under Z-score, a gene measured in only one of the selected samples has no standard deviation and disappears from the figure.
- A single replicate per condition is not a problem here — nothing on this page is a test — but the z-score is then taken across conditions rather than within them.
- Selecting Imputed Data draws imputed values in the same colours as measured ones, with nothing to distinguish them.

4 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
db = load_go_db()
fig = proteins_in_pathway(data, meta, db, "proteasome complex")
```

R

```
db <- load_go_db()
fig <- proteins_in_pathway(data, meta, db, "proteasome complex")
```

The term list is not a menu here: load_go_db returns the bundled table and the term is named to proteins_in_pathway as a string, matched against its description, so any of the 22,074 terms

can be drawn without an enrichment result in front of it. The defaults line up with the page — value_type="z", remove_empty=True, both clustering switches off, every condition drawn when conditions is left unset — with one exception: id is on by default and puts the sample identifier under each column label, which is **Toggle ID**, and that is off on the page.

5 Exporting

The figure is rendered at the width, height and resolution set on the page, which is why those three controls do not change what is on screen: the displayed image is fitted to the window, the downloaded one to the numbers. Raise the height for a term of many genes before turning **Show gene labels** off — a taller figure keeps the labels readable where a wider one does not. Open Image gives the PNG.

6 Further reading

Gene Ontology: enrichment analysis distinguishes an enrichment result from a list of annotated genes. A heatmap visualizes values for such a list; it does not itself test activation of the term.

Behind this page

Function	What it produces
proteins_in_pathway	The heatmap of one term's genes across samples
load_go_db	The bundled Gene Ontology term-to-gene table the term list is read from